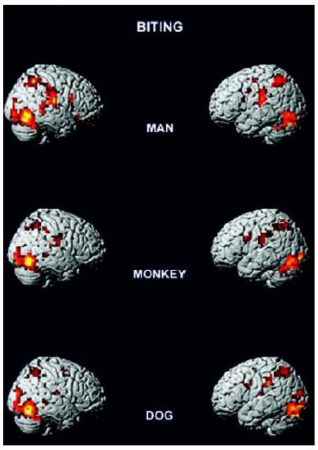
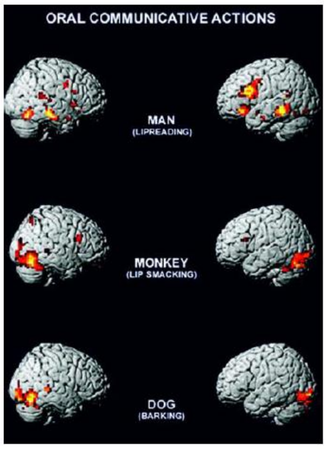




In a comparison between capoeira dancers and classical ballet dancers both were danced while presented both some video, but mirror neurons were responding higher when each of them were shown videos of people dancing their style.

This is very important in order to actually understand how motor information is really rooted in my perception, and how it allows me to predict in advance, and also the least I know some motor condition the less I'm able to predict that specific behaviour. When you see and are familiar with an action, while I'm doing that and someone is watching if I would record their muscle, they would be ready to enact the movement, and this is also a strategy used for therapy, exposing you to movements that you are not able to perform anymore, in order to have your motor system to get ready to represent and retrieve that kind of motor action. Also, the context of an action can tells us how we can modify.

Mirror neurons too are present in blind people, so people that have never seen are able to have mirror neurons system responding to the recognition of action, and even their mirror neurons system was responding more to action they were familiar with.

Mirror neurons appear to be a sort of like of empathic too; if I'm able to enter other people shoes and understand their intentions, I'm also able to understand their feelings and when you look at regions that are involved in emphatic you see that is plenty of regions that are present in the mirror neurons system.

Anthropomorphism is a feature you can play with in your device; in order to build something that your brain is able to embody in your body schema, because we recognize it most.

These aspects will be fundamental if you are going to look at this in terms of embodied cognition; the idea that every time you have to perform an action, every time we have to process a stimulus or try to have a broader conceptual representation, this is rooted somehow in a sensory motor experience. How can I work with a tool or device to be embodied with the patient, to be truly part of his body representation, part of his motor repertoire.

Plasticity

Plasticity has been defined as an intrinsic property of the human brain, representing a sort of invention of evolution that enables the nervous system to overcome the constraints imposed by its genome, allowing it to adapt to the environment or physiological changes and experience.

We often think of plasticity as something that is beneficial, something that even in the condition of damage and lesion could help recovering, but this is not often the case; we also have the case of maladaptive plasticity, some kind of adaptation to a lesion that is not perfectly fitted with the means of the brain, and more importantly how is the brain able to adapt to an equation in which we add multiple limbs?

Plasticity is the ability of neural circuits to modify their structure and function in response to stimuli, both during development and throughout adult life. During the early stages of brain development, plasticity is extremely high: some neuronal circuits are selected while others are eliminated; in the first months of life how brain matures, it takes 10-18 months to learn how to talk and walk and we are mostly blind at the birth, so our brain has to adapt. In adulthood, many circuits remain largely stable, yet neuronal populations continue to exhibit a certain degree of dynamism, reorganizing under the influence of the external environment to meet specific motor, sensory, cognitive, or emotional demands. There is a period of life, until 30-33 years, where the brain is adapting to the effect of hormonal changes (pruning). Our brain remains stable until 60-65 years and then will start facing issue of cognitive decline, nonetheless the ability to change in time is not the same for all cognitive motor perceptual domains, so we have different windows through which this kind of changes can occur more easily.

Development of the embryonic brain:

- CNS as “a fluid filled tube”.
- Proliferation and synaptogenesis.
- Migration.
- Differentiation and maturation.
- Myelination.
- Excess production of neurons.
- Development of connectivity.
- NGF and other neurotrophins.
- Pruning.

The migration of cells gets to the development of the brain birth; the brain of a newborn is not proportional with the rest of the body. This is all regulated at molecular level since we have molecules called neurotrophic, that are feeding the neurons and the synapsis to grow, and this phase is really important. But there is an intrinsic organization that is determined at the genetic level, the brain is already defined in a specific way innately (even blind brain has a large scale of organization very similar to the sighted ones, it’s not just the motor experience).

Following several cell migration steps, three main layers emerge:

- **Mesoderm** cells give rise to the structures including muscles, bones, cartilage and the vascular system.
- **Epidermal ectoderm** cells give rise to structures including the skin, nails and sweat glands.
- **Neuroectoderm cells** give rise to the brain and the nervous system.

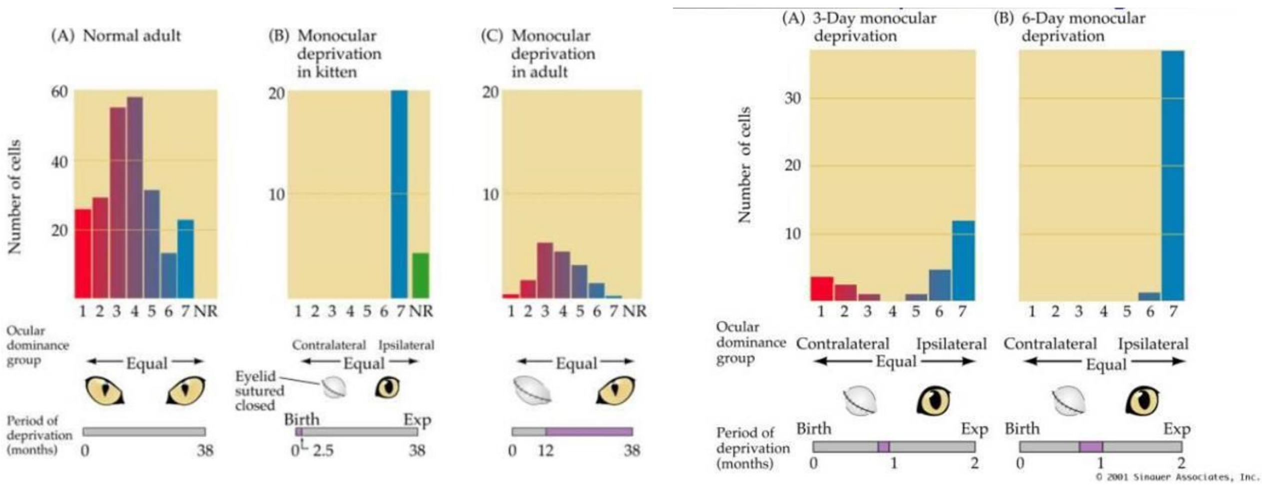
At first it starts as a plate, then it closes into a tube, and then we start having areas and portion that appear to have and rely on these projectile cells that create the whole brain. (NOT GONNA ASK ABOUT BRAIN DEVELOPMENT, TO READ MORE SEE THE SLIDES).

We define as sensitive period, a period that is a limited epoch during development in which the effects of experience on the brain are usually stronger; certain capacities are quickly shaped or altered by experience. We have a specifical architecture of the brain, but there are periods in development in which experience shape more the functional and structural organization of the brain, and these are called sensitive periods.

Among the sensitive periods we define a class called critical periods, that results in irreversible changes in brain functions. After the end of the critical period, effects of atypical experience during this period cannot be remediated by restoring typical experience. Critical periods are primarily related to perinatal time, so after conception we have several critical periods, first for sensory motor, olfactory, taste and auditory and vision. During these critical periods, on of these environmental input, for instance a sensory modality like vision is missing, we are not able to fully restore all functions if in the future, for instance, I'm recovering vision if we treat the problem at the base, because some structural functional features cannot be recovered anymore since the critical period is over. This is why most of the ways of restoring vision are typically done in people that get blind late, and also why when we have a baby born deaf, we try to implant the cochlear implant as soon as possible, because we don't want to miss the critical period.

Sensitive critical periods are reflected in behaviour, but they are property of neural circuits. It means that critical periods reflect for instance the ability to develop dominance of the eyes, develop acuity so on, and this is something that we actually reflect behaviourally. But critical period is a period where we have higher development of synapsis and higher metabolism, so there are always structural components that subserve this development properties.

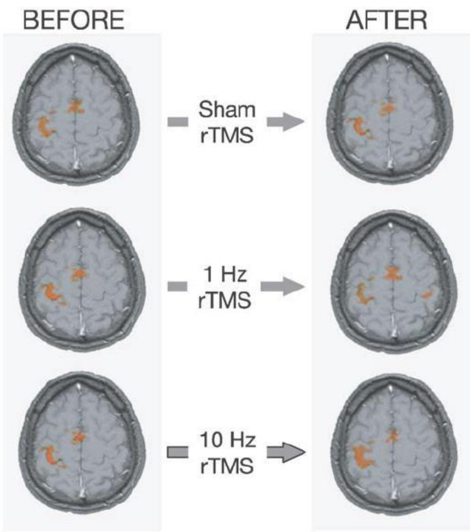
If you deprive the monkey at the beginning of their life from 2 weeks to 18 months of age (first 4-5 years for a monkey), and you look at the visual cortex, you see that the ocular dominance is lost, so we have one eye that appears to dominate, that is the non-deprived eye.



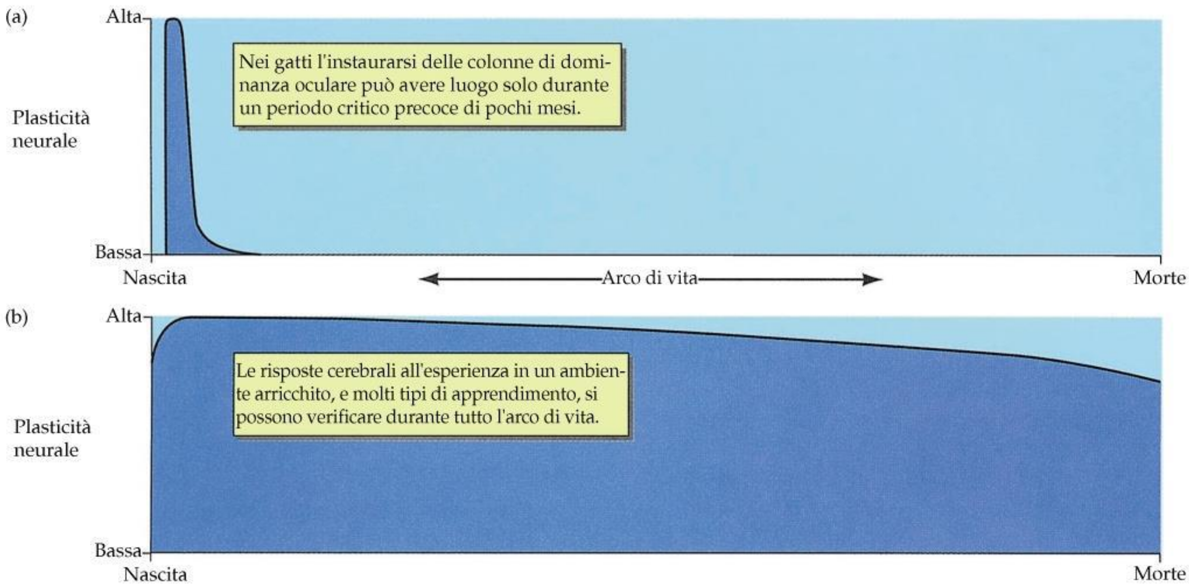
When you have a monocular deprivation, you clearly have a number of cells responding only to the eye that is not been damaged. If you deprive the adult, you see that the change that impacts is reducing the number of cells responding to the stimulus, but not the sensitive distribution of the left or the right eye. So, one of the things is that the impact on changes that could occur in the brain is different according to the age these vision deprivations occur.

This ideal plasticity is also related to time, because if we are not actually defining what we refer to as plasticity, and we refer to plasticity as a wide gamut of phenomena, even the modulation with repetitive transparent magnet stimulation could induce very swift changes in few seconds.

If you stimulate motor activity for different time and frequency, you see in few seconds an increase in lateral connectivity.



Plasticity may take much longer, so time window is an issue, an example is relative to the exogenous modulation of plasticity, thinking about the hormonal effects in brain structure, and the difference between male and female. Male has a much more specific element on one side and female has more from a lateral. So, one thing clearly evident is that there are specific aspects that appear just at a very short period of time, and these are the critical sensitive periods, some other form of plasticity, like learning, last for the whole life, we can just have a slight reduction but we are able to learn things along most of our lives.



We could have aspects of periodic learning, and this is not often present in humans but very common in animal for instance. There are birds that remember up to 80 sites in which they hide food, and each summer they have to change and remember other 80.

There are domains instead in humans too, that have a very specific learning window, and this is typical for language. If you want to learn language early years of life are the best, because we are more able to separate different rules of languages. This ability remains for a long period of life, and you still use language related brain areas to learn new language, but after that age you can start a new language, but you aren't using anymore this area, but you are using memory.

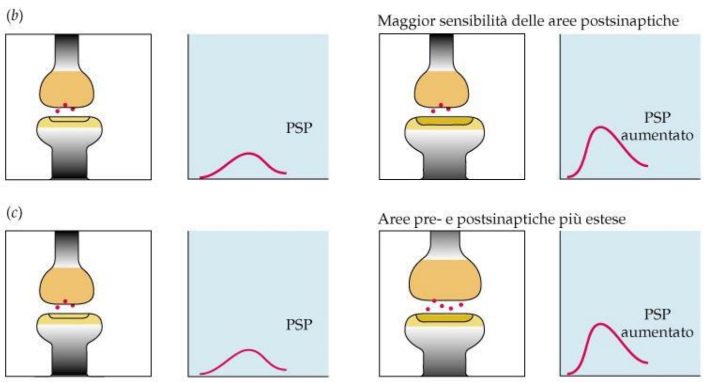
These aspect of learning are something we can define as permanent changes in behaviour produced by experience. Learning involves changes in the nervous system that are produced by experience, the more you experience and the highest the changes in the nervous system. Changes in the nervous system are physical, and learning is what allows us to adapt to the environment.

Our psychology strength decides to define different way of learning, like a perceptual learning, stimulus response learning (conditioning, performing an action in response to a specific stimulus), motor learning (formation of new circuits within the motor system), relational learning (identifying relationships or connections among stimuli).

In musicians the anterior commissure, that relates to connection between the two motor cortices, is thicker and if we have to look specifically at the differences between the two motor cortices you see that while is higher for non-musicians, musicians tend to have a very similar size of the two motor cortices, they have a great coordination of different body parts. Typically, this kind of phenomenon occur somehow twice in life. These changes are higher the earliest in life.

The anatomical differentiation of the brain is clearly tight to experience. Brain maturation has two specific moments, one in early life and the other one is in adolescence; if in early life we have these aspects of synaptogenesis, during adolescence we have a phenomenon that is called pruning. This is a phenomenon that tells us "Use it or lose it", since during adolescence we have actually to select which are our skills and interest, and the brain while is growing modulated by hormones is also adapting to changing environment. Brain cells that are typically strengthen are showing as morphologically as enriched or impoverished, and this kind of thing comes from experiment like the one in the slide with the rats. In an enriched environment you have more animals, it's crowded and not very nice, but at least you have social interaction, very close to real life. These aspects are typically impacting the size of the brain; one of the effects was the one described in the musicians and one other important study is the one conducted on London taxi drivers. In London you have to learn different streets name and indeed their posterior hippocampus, related to long term memory related to navigation, gets broader and bigger compared to ordinary people.

Our idea is also how can neurons actually try to compensate when we have a damage and we may have formation of new synapsis, so we are extending the map of interest. This is called typically sprouting. We can also have a sort of hypersensitivity induced by inactivity of the innervation, and the size of the synapsis increases.



We already talked about cross-modal organization, with early visual cortex in blind people that is responding to non-visual stimuli, so following sensory deprivation primary and secondary sensory areas indicated to that perceptual modality, reorganize to respond to other sensory modalities. We still have a hill defined understanding of the functional role of this; we can just regard that visual cortex starts to respond to non-visual stimuli.

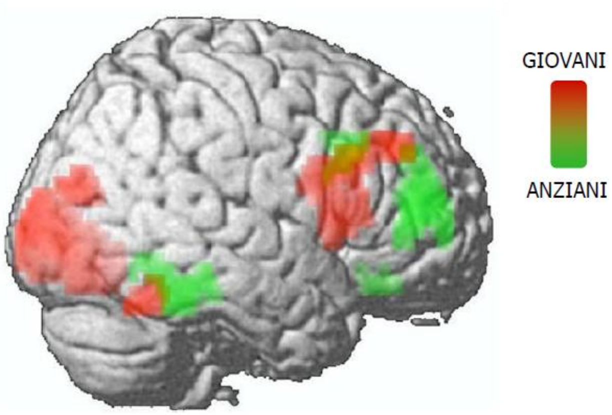
We then have compensatory masking: following injury or learning, an implicit or explicit change in task-execution strategy occurs, allowing a secondary cognitive process—normally playing a minor role—to assume a primary role). The idea is that when there is something difficult, I'm going to rely on other processing in order to achieve that specific task.

Expansion of the functional map: following injury, a functional region adjacent to the damaged area expands its functional/neural territory into neighbouring cortex. Learning can also induce an enlargement of the functional extension of a region into adjacent cortical areas.

Adaptation of homologous regions: following injury, functionally homologous regions in the contralateral hemisphere can compensate for the lost functions. This is something we are actually observing in aging.

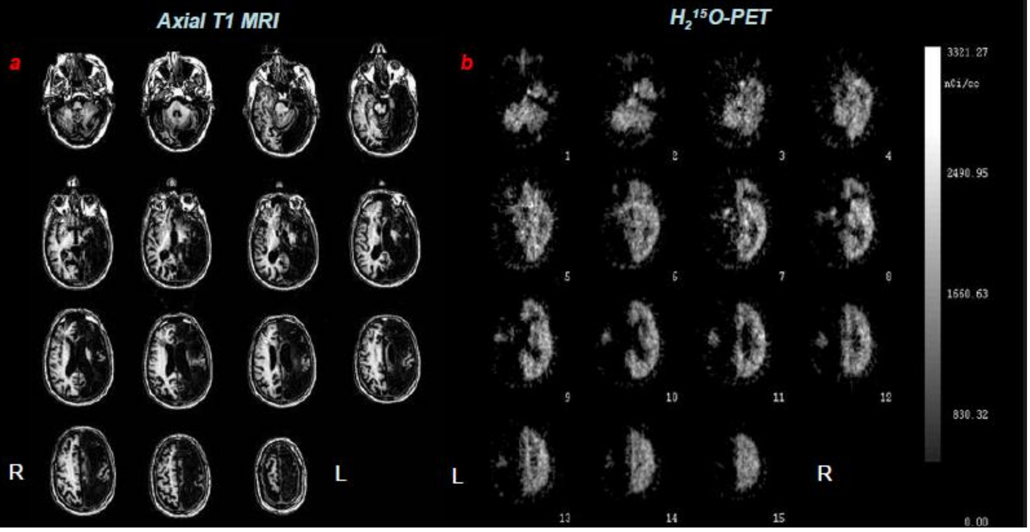
In a task there can be old people performing and old people not performing on it, and they can be called old high and old low. Old low try to extend the map but are not performing well, and in order to compensate they have a recruitment of the homologous regions in the contralateral hemisphere.

During aging you can have different effects also, young and older people are actually even recruiting additional regions involving a single function, and this is called compensatory masking; to do the same thing you recruit other regions. For example, you present faces and then you maintain that information for a few seconds and then you ask to recognize which was the first one; to do that both old and young people are recruiting the visual cortex and the right prefrontal cortex, but also older are actually extending the map for recruiting not just that portion of the prefrontal cortex but also another one, also if you give them drugs that potentiate memory you see that they act in different regions in the two. This modulation tells us that this compensation that is truly effective is the effect of the drug, so younger and older people modulate different portions.



PET and fMRI in the Study of Functional Plasticity in the Human Brain

A 67 y/o right-handed male survived a **massive stroke of the left hemisphere** at age 47 yr. In the left hemisphere, structure (a) and metabolic activity (b) were preserved in limited areas only. We studied **potential compensatory effect** of the R hemisphere (brain plasticity) in carrying out activities normally sustained by the L hemisphere.

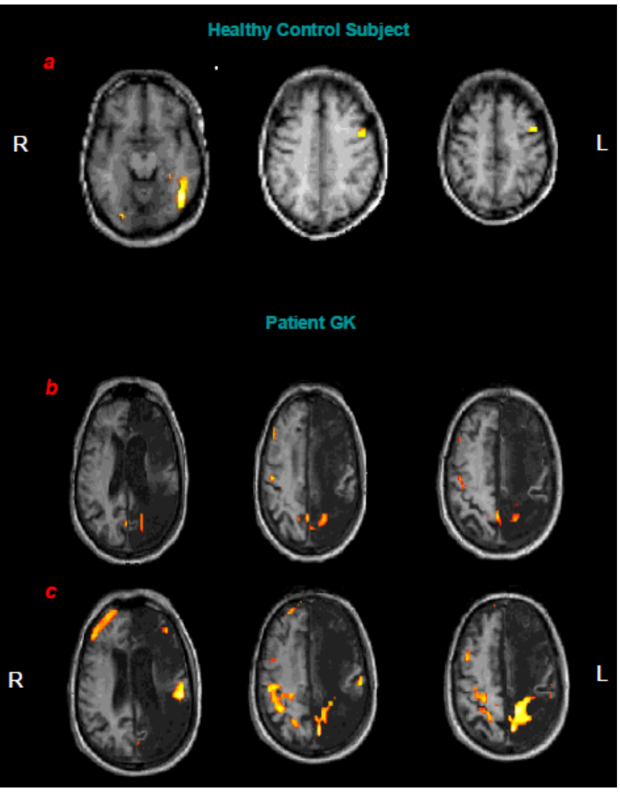


Normally mathematical functions rely on parietal and frontal areas of the left hemisphere (a).

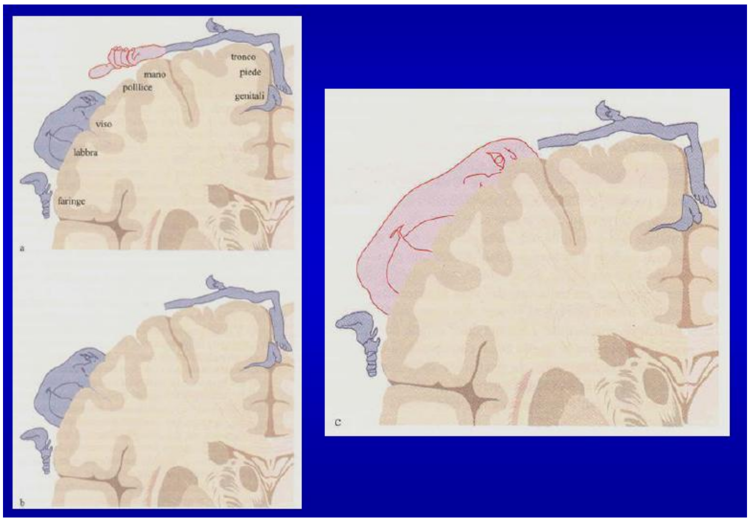
In patient GK performance on mathematical tasks was highly compromised and associated with limited activation of residual parietal areas in the left hemisphere (b).

After a period of training the patient reacquired some ability to perform simple mathematical operations. Improved mathematical performance was associated with activation of homologous parietal and frontal areas in the right hemisphere (c).

These findings indicate a compensatory process through functional plasticity in the adult brain.



As for the Ramachandran's procedure, since mostly form a sensorial component, you have the expansion in the sensory homunculus of the face perception area. While we are having novel synapsis, if you touch the patient in the cheek amputees can report that you are touching their head, and this is because brain is not properly localizing cortically the location of the stimulus.



The take on message is that we may describe different forms of the way the brain prefers and tries to accommodate, even if we are not fully understanding the principles, we have understood that this capacity of the brain to adapt to environmental pressure changes in time, changes according to the specific domain that you are exploring, and we have no specific way of classifying how well this mechanism is actually compensatory when there is damage. This is why several researchers are really wondering whether you have plastic phenomena in specific conditions and situations, and this is also primarily relating to the big trade-off between what is innately present in birth in the nervous system, and which are the features that you can modify during life according to environmental experience.

We have to remember that when we change a function often there is a change in the structure of the brain, or you can describe the change in the function but in order to stabilize it you need to have a structural change too. Secondly, with several tools we can describe changes in behaviour or at the broad level but never forget that these changes occur at the molecular level.

We are aware that these maturation skills are impacted by stress, drugs and neurological disorders, and we are yet to understand how we can foster and promote this kind of restoration.